

Assignment 3: Variational Formulations and Energy Principles

ME/BME 736 — Advanced Finite Element Methods

Prashant K. Jha

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Show the main steps in each derivation. This assignment concludes the continuous mathematical and variational foundations developed in Chapters 2–4. No finite element discretization is required.

Problem 1 (25 marks): Applying the Lax–Milgram theorem to diffusion with a zero-order term

For help: See Chapter 3, Section *Well-posedness and stability*. For the definitions of continuity and coercivity and for the Cauchy–Schwarz and trace inequalities, see Chapter 2, Sections *Continuity, coercivity, and well-posedness* and *Reference inequalities*.

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain with

$$\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}, \quad \Gamma_D \cap \Gamma_N = \emptyset,$$

where Γ_D has positive boundary measure. Consider

$$-\nabla \cdot (k \nabla u) + cu = f \quad \text{in } \Omega,$$

with

$$u = 0 \quad \text{on } \Gamma_D,$$

Define the diffusion-flux vector and its outward normal component by

$$\mathbf{q} = -k \nabla u, \quad q = \mathbf{q} \cdot \mathbf{n}.$$

The boundary conditions are

$$\mathbf{q} \cdot \mathbf{n} = q \quad \text{on } \Gamma_N.$$

The additional term cu contains no derivative of u and is therefore a **zero-order term**. In diffusion models it is also commonly called a reaction term.

Assume

$$k, c \in L^\infty(\Omega), \quad 0 < k_0 \leq k(\mathbf{x}) \leq k_1, \quad 0 < c_0 \leq c(\mathbf{x}) \leq c_1$$

almost everywhere in Ω , with

$$f \in L^2(\Omega), \quad q \in L^2(\Gamma_N).$$

Use

$$V = H_{0,\Gamma_D}^1(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

1. Starting from the strong problem, derive

$$a(u, v) = \ell(v) \quad \forall v \in V,$$

and identify $a(w, v)$ and $\ell(v)$.

2. You may use the estimates

$$|a(w, v)| \leq (k_1 + c_1) \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)},$$

$$a(v, v) \geq \min\{k_0, c_0\} \|v\|_{H^1(\Omega)}^2,$$

and

$$|\ell(v)| \leq \left(\|f\|_{L^2(\Omega)} + C_{\text{tr}} \|q\|_{L^2(\Gamma_N)} \right) \|v\|_{H^1(\Omega)}.$$

Explain which hypotheses of the Lax–Milgram theorem these estimates establish and state the conclusion of the theorem for this problem.

Problem 2 (20 marks): From an energy functional to the variational problem

For help: See Chapter 3, Section *Energy minimization and first variation*. For directional derivatives and first variations, see Chapter 2, Section *Directional derivatives and first variations*.

Use the same coefficient functions k and c , source f , outward normal flux q , and boundary partition as in Problem 1. Now allow a prescribed value

$$u = g \quad \text{on } \Gamma_D,$$

and define

$$\mathcal{U}_g = \{z \in H^1(\Omega) : z = g \text{ on } \Gamma_D\}, \quad \mathcal{V}_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

For $z \in \mathcal{U}_g$, consider

$$\Pi(z) = \frac{1}{2} \int_{\Omega} (k|\nabla z|^2 + cz^2) \, dx - \int_{\Omega} fz \, dx + \int_{\Gamma_N} qz \, ds.$$

1. Let $u \in \mathcal{U}_g$. A variation, or perturbation, of u should preserve the prescribed value g on Γ_D . Explain why choosing $v \in \mathcal{V}_0$ ensures that

$$u + \varepsilon v \in \mathcal{U}_g$$

for every scalar ε .

2. Compute the first variation

$$\delta\Pi(u; v) = \left. \frac{d}{d\varepsilon} \Pi(u + \varepsilon v) \right|_{\varepsilon=0}.$$

3. Set

$$\delta\Pi(u; v) = 0 \quad \forall v \in \mathcal{V}_0.$$

Show that the resulting equation can be written as

$$a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0,$$

where the bilinear form a and linear functional ℓ are exactly the same forms identified in Problem 1.

Problem 3 (30 marks): From strong form to variational form in linear elasticity

For help: Review Chapter 3, Section *Canonical scalar diffusion problem*, especially *Weighted residual and integration by parts*, *Trial and test spaces*, and *Variational problem*. For the corresponding elasticity model, see Chapter 4, Section *Small-strain elasticity*.

Let an elastic body occupy a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$, where $d = 2$ or 3 . Its boundary is divided into two non-overlapping parts,

$$\partial\Omega = \overline{\Gamma_u} \cup \overline{\Gamma_t}, \quad \Gamma_u \cap \Gamma_t = \emptyset.$$

The unknown displacement field is

$$\mathbf{u} : \Omega \rightarrow \mathbb{R}^d.$$

The quasistatic equilibrium equation is

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{b} \quad \text{in } \Omega,$$

with boundary conditions

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_u,$$

and

$$\boldsymbol{\sigma} \mathbf{n} = \bar{\mathbf{t}} \quad \text{on } \Gamma_t.$$

Here:

- $\mathbf{u}$ is the displacement field;
- $\bar{\mathbf{u}}$ is the prescribed displacement on Γ_u ;
- $\boldsymbol{\sigma}$ is the Cauchy stress tensor;
- $\mathbf{n}$ is the outward unit normal to the boundary;
- $\bar{\mathbf{t}}$ is the prescribed traction vector, or force per unit area, on Γ_t ;
- ρ is the mass density;
- $\mathbf{b}$ is the body force per unit mass, such as gravitational acceleration;
- therefore $\rho \mathbf{b}$ is the body force per unit volume.

For small deformations, the infinitesimal strain tensor is

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T).$$

For an isotropic linear elastic material,

$$\boldsymbol{\sigma}(\mathbf{u}) = 2\mu \boldsymbol{\varepsilon}(\mathbf{u}) + \lambda (\nabla \cdot \mathbf{u}) \mathbf{I},$$

where λ and μ are the Lamé material parameters and $\mathbf{I}$ is the identity tensor. The stress tensor $\boldsymbol{\sigma}$ is symmetric.

Begin with a smooth vector test function $\mathbf{v}$ satisfying

$$\mathbf{v} = 0 \quad \text{on } \Gamma_u.$$

You may use the tensor integration-by-parts identity

$$\int_{\Omega} (-\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} \, dx = \int_{\Omega} \boldsymbol{\sigma} : \nabla \mathbf{v} \, dx - \int_{\partial\Omega} (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{v} \, ds.$$

Because $\boldsymbol{\sigma}$ is symmetric,

$$\boldsymbol{\sigma} : \nabla \mathbf{v} = \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{v}),$$

and for isotropic linear elasticity,

$$\boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) = 2\mu \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) + \lambda (\nabla \cdot \mathbf{u})(\nabla \cdot \mathbf{v}).$$

1. Multiply the equilibrium equation by $\mathbf{v}$, integrate over Ω , and use the identities above together with the boundary conditions to obtain

$$\int_{\Omega} [2\mu \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) + \lambda (\nabla \cdot \mathbf{u})(\nabla \cdot \mathbf{v})] \, dx = \int_{\Omega} \rho \mathbf{b} \cdot \mathbf{v} \, dx + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v} \, ds.$$

2. Assume

$$\lambda, \mu \in L^\infty(\Omega), \quad \rho \mathbf{b} \in L^2(\Omega; \mathbb{R}^d), \quad \bar{\mathbf{t}} \in L^2(\Gamma_t; \mathbb{R}^d),$$

and assume that the prescribed displacement $\bar{\mathbf{u}}$ is the boundary value of a function in $H^1(\Omega; \mathbb{R}^d)$. Following the same reasoning used for the scalar diffusion problem in Chapter 3, identify appropriate trial and test spaces for $\mathbf{u}$ and $\mathbf{v}$ so that all terms in the integral identity are well defined.

3. Define the bilinear form $a(\mathbf{u}, \mathbf{v})$ and the linear functional $\ell(\mathbf{v})$, and write the complete variational problem in the form

$$\text{find } \mathbf{u} \in \mathcal{U}_{\bar{\mathbf{u}}} \text{ such that } \quad a(\mathbf{u}, \mathbf{v}) = \ell(\mathbf{v}) \quad \forall \mathbf{v} \in \mathcal{V}_0.$$

Problem 4 (25 marks): A manufactured solution in linear elasticity

For help: See Chapter 4, Section *Small-strain elasticity*, especially *Infinitesimal strain*, *Stress*, *traction*, and *balance of momentum*, and *Linear elastic constitutive response*. For tensor notation and tensor-vector multiplication, see Chapter 2, Section *Scalars, vectors, and tensors*.

A manufactured solution starts from a chosen displacement field and determines the body force and boundary tractions for which that displacement is an exact solution.

Let

$$\Omega = (0, L) \times (0, H) \subset \mathbb{R}^2,$$

and consider isotropic small-strain linear elasticity with

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T),$$

and

$$\boldsymbol{\sigma}(\mathbf{u}) = 2\mu \boldsymbol{\varepsilon}(\mathbf{u}) + \lambda \operatorname{tr}(\boldsymbol{\varepsilon}(\mathbf{u})) \mathbf{I},$$

where $\lambda > 0$ and $\mu > 0$ are material constants and $\mathbf{I}$ is the 2×2 identity tensor.

Consider the displacement field

$$\mathbf{u}(x_1, x_2) = \begin{bmatrix} \alpha x_1^2 \\ 0 \end{bmatrix},$$

where α is a prescribed constant. Let the left edge $x_1 = 0$ be a displacement boundary with $\mathbf{u} = 0$, and treat the remaining three edges as traction boundaries.

1. Compute

$$\nabla \mathbf{u}, \quad \boldsymbol{\varepsilon}(\mathbf{u}), \quad \operatorname{tr}(\boldsymbol{\varepsilon}(\mathbf{u})), \quad \boldsymbol{\sigma}(\mathbf{u}).$$

2. Compute

$$-\nabla \cdot \boldsymbol{\sigma}(\mathbf{u})$$

and hence determine the body-force field $\rho \mathbf{b}$ for which the prescribed displacement is an exact equilibrium solution.

3. Determine the outward unit normal on the right, top, and bottom edges and compute the traction

$$\bar{\mathbf{t}} = \boldsymbol{\sigma}(\mathbf{u})\mathbf{n}$$

on each edge.